

Malek Bsaissa

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Education

ESPRIT – School of Engineering 2022 – 2027 (Expected)
Engineering Degree in Computer Science / *Specialty: ARCTIC (Cloud Computing & DevOps)*
Key coursework: Software Engineering, Data Structures & Algorithms, OOP, Web Development, Cloud Infrastructure

Technical Skills

Languages	Java, Python, JavaScript/TypeScript, C/C++, PHP, SQL
Frameworks	Spring Boot 3, Angular 18, React 18, Laravel 12, JavaFX, Symfony, Qt
Cloud & DevOps	Docker, Kubernetes, Ansible, OpenStack, Nginx, Karmada, Azure AKS
AI / ML	LLM integration (OpenAI, Groq, Gemini), Python (pandas, scikit-learn)
DB & Messaging	MySQL, Firebase, WebSocket/STOMP, REST APIs
Tools	Git, Maven, Linux, CesiumJS, Stripe API, Node.js

Experience

Full-Stack Developer Intern Jun – Aug 2025
Smart Skills | El Ghazela Technology Park, Ariana

- Engineered **MySkills**, a multi-role training management platform (Admin, Coordinator, Trainer, Trainee) with **Laravel 12**, **React 18.3**, and **MySQL**
- Built 70+ RESTful API endpoints with token-based authentication via **Laravel Sanctum** and granular **RBAC** enforced end-to-end across back end and front end
- Delivered an automated certificate generation pipeline (**Intervention Image** + **ImgBB API**) and a dual-channel notification system (in-app + SMTP email)

Operations Intern Jun 2023
Tunisair | Tunis-Carthage International Airport
Observed and supported airline ticketing, refund processing, and day-to-day operational workflows.

Projects

Elif – Pet-Care Platform with Multi-Cluster Deployment 4th Year | Team Project

- Architected and built the **Community module** — Reddit-style communities, threaded forum with hot/new/top ranking, and real-time 1-on-1 WebSocket/STOMP chat with AI-powered summarization (Groq/Gemini) — using **Angular 18**, **Spring Boot 3**, and **MySQL**
- Provisioned a private **OpenStack** cloud and deployed a production **Kubernetes** cluster with **Ansible**; orchestrated multi-cluster workloads across on-prem and **Azure AKS** via **Karmada**
- Implemented HA infrastructure: **NGINX Ingress**, **MetalLB**, **kube-vip**, and automated cron backups (mysqldump, rsync, etcd snapshots)

AtlasMesh – Real-Time 3D Geospatial Dashboard Personal

- Aggregated 18 live data sources (aircraft, ships, ISS, Starlink, earthquakes, volcanoes, wildfires, weather) and rendered them on a **CesiumJS** 3D globe
- Engineered real-time WebSocket diff-based updates with **Node.js** + **TypeScript** and **SGP4** orbital propagation for satellite trajectory computation
- Applied occlusion culling, entity clustering, and fuzzy search across 15 000+ live entities — deployed at atlasmesh.onrender.com

The 12th Man – ML-Powered Football Predictions Academic

- Predicted Premier League standings, match outcomes, and emerging player talent using **Python** (pandas, numpy, scikit-learn) with an integrated web front end

- Applied feature engineering on multi-season match data; evaluated models (Logistic Regression, Random Forest, XGBoost) against historical baselines

WamiaGo – Transportation Management Platform

3rd Year

- Built the **electric bicycle management module** in a **JavaFX/MySQL** desktop app, integrating **OpenAI** and **Azure Speech** APIs for AI-assisted ride planning, QR-based rentals, and real-time tracking
- Developed a companion **Symfony/PHP** web platform sharing data models and business logic with the desktop application for seamless cross-platform operation

Ripple Insurance System – IoT Accident Detection

2nd Year

- Simulated real-time accident detection using **C++**, **Qt**, and live **Arduino** sensor integration with automated alert dispatch

Languages

Arabic	Native
English	Advanced — professional proficiency
French	Intermediate (B1)